MIL-I-27882A(USAF
18 November 1963
Superseding
MIL-I-27882 (USAF)
18 June 1962

MILITARY SPECIFICATION

INDICATOR, LIQUID OXYGEN QUANTITY GMU-37/A

1. SCOPE

1.1 This specification covers one type of liquid oxygen quantity indicator designated Type GMU-37/A. The indicator is of the capacitance type and indicates from 0 to 5 liters of liquid oxygen.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specification for
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-831	Test Reports, Preparation of
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6680

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3. REQUIREMENTS

3.1 QUALIFICATION.- The indicator furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 GENERAL SPECIFICATION.- The requirements of MIL-I-25645 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 CONFIGURATION.- The indicator shall consist of a single unit containing all of the equipment necessary to meet the performance requirements specified herein and in MIL-I-25645.

3.4 CASE.- The length of the case shall not exceed 5.0 inches.

3.5 DIAL.- The dial for the indicator shall conform to figure 1.

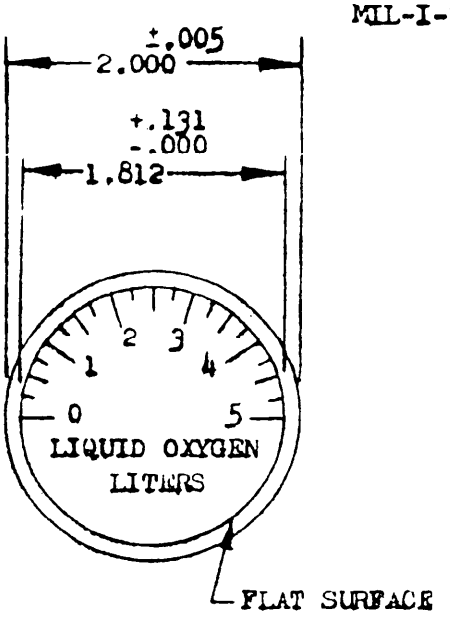
3.6 ELECTRICAL CONNECTOR.- The electrical connector shall be in accordance with figure 2. The indicator shall be designed internally so that it will function properly when the mating electrical connector is wired as shown on figure 2.

3.7 CAPACITANCE.- For design purposes, the capacitance for zero reading shall be 63.5 micromicrofarad (uuf).

3.8 ADJUSTMENTS.- The zero calibration adjustment for the indicator shall be capable of adjusting the pointer to zero indication over an input capacitance range of at least 60.0 uuf to 67.0 uuf. With the indicator adjusted to zero indication at 60.0 uuf input capacitance, the indicator shall be capable of calibration to full scale indication at 89.0 uuf input capacitance by means of the full calibration adjustment. With the indicator adjusted to zero indication at 67.0 uuf input capacitance, the indicator shall be capable of calibration to full scale indication at 96.0 uuf input capacitance by means of the full calibration adjustment.

3.9 WEIGHT.- The indicator shall weigh not more than 1.5 pounds.

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NUMERALS AND LETTERS SHALL CONFORM TO MS33558

MARKINGS	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005	FINISH COLOR PER FED-STD-595
NUMERALS	.140	.025	37875
LIQUID OXYGEN	.140	.020	
MAJOR GRAD.	.188	.031	
MINOR GRAD.	.125	.015	
SUB MINOR	.080	.015	
LITERS	.100	.015	
BACKGROUND OF DIAL			37038

DIMENSIONS IN INCHES

FIGURE 1 - DIAL

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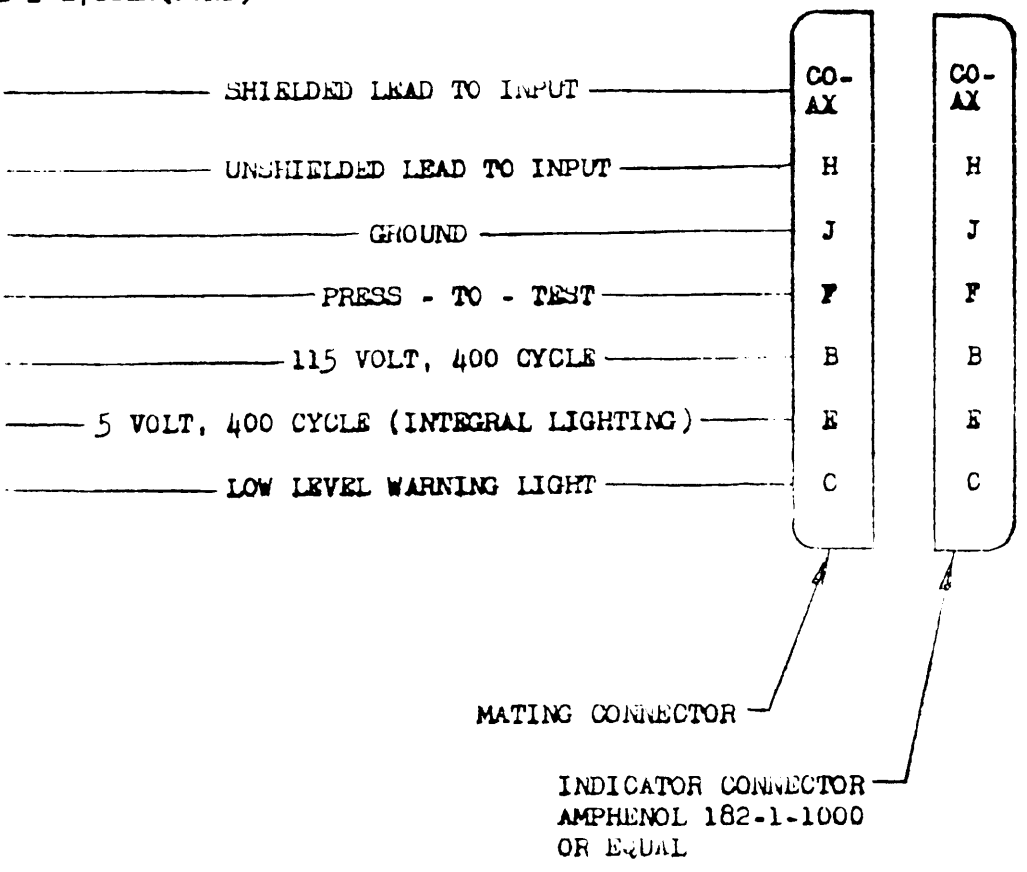


FIGURE 2 - CONNECTOR WIRING

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3.10 ACCURACY.- When the indicator set is tested in accordance with MIL-I-25645, the accuracy of the indicator shall be in accordance with table I. The tolerance for room-temperature readings is the allowable deviation from the nominal calculated values. All other tolerances are the allowable deviations from the actual room-temperature readings.

TABLE I - TOLERANCE FOR CAPACITANCE READINGS

Test	Tolerance Plus or Minus
Accuracy - room temperature	0.4 uuf
Low temperature operation	0.5 uuf
High temperature operation	0.5 uuf
Vibration	0.3 uuf
Power supply variation	0.3 uuf
Low temperature storage and operation	0.5 uuf
High temperature storage and operation	0.5 uuf
Salt spray	0.3 uuf
Cycling	0.3 uuf

3.11 POWER SUPPLY VARIATION.- Variations of supply voltage from 105 volts to 125 volts and frequencies of 320 to 480 cycles per second shall not cause indicator errors exceeding those specified in table I when the indicator is tested as specified in MIL-I-25645. For the 105-volt test point, the indicator shall be checked at frequencies of 320 and 480 cycles per second. For the 125-volt test point, the indicator shall be checked at frequencies of 320 and 480 cycles per second.

3.12 INTEGRAL LIGHTING.- The indicator shall have integral lighting in accordance with MIL-L-27160.

4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 SAMPLING, INSPECTING, AND TESTING.- Sampling, inspecting, and testing of indicators shall be in accordance with MIL-I-25645.

4.3 QUALIFICATION TESTING

4.3.1 QUALIFICATION TEST SAMPLE TESTED BY THE CONTRACTOR.- The contractor shall subject one indicator to the qualification tests specified in 4.3.4.

4.3.2 QUALIFICATION TEST REPORT.- After the contractor completes the qualification tests, he shall prepare a qualification test report in accordance with MIL-STD-831 and shall furnish three complete copies of the report to the qualifying activity.

4.3.3 QUALIFICATION TEST SAMPLE FOR THE QUALIFYING ACTIVITY.- Along with the qualification test report, the contractor shall submit the indicator that was tested as specified in 4.3.1.

4.3.4 QUALIFICATION TESTS.- Qualification tests shall consist of all tests described under Test Methods of MIL-I-25645.

4.3.5 QUALIFICATION RELIABILITY TESTS.- One qualification sample of the equipment shall be tested by the contractor for reliability and shall be continued through a longevity test in accordance with MIL-R-26667. Duration of the mean cycles between failures (MCBF) demonstration test on each unit shall not be less than three times the specified MCBF. Recording, data handling, and reporting procedures shall be in accordance with MIL-R-26667.

5. PREPARATION FOR DELIVERY

5.1 The indicator shall be prepared for delivery in accordance with MIL-I-25645.

6. NOTES

6.1 INTENDED USE.- The indicator covered by this specification is intended for use in indicating the quantity of liquid oxygen remaining in converters in F-5A and C-142 airplanes.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Levels of preparation for delivery required.

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6.3 QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Systems Engineering Group, Attention: SEMES, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

Form 1426

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